Chapter 8: Electromagnetic Waves

Formula Sheet + Tips & Tricks

Formula Sheet

I. Displacement Current

Formula: $I_d = \varepsilon_0 A \dfrac{dE}{dt}$

I_d = displacement current, ε_0 = permittivity of free space, A = plate area, $\dfrac{dE}{dt}$ = rate of change of electric field.

II. Electromagnetic Waves

Formula (Speed in a medium): $v = \dfrac{c}{\sqrt{\mu_r \varepsilon_r}}$

v = speed of EM wave in medium, c = speed of light in vacuum, μ_r = relative permeability, ε_r = relative permittivity.

Formula (Wave equation): $c = f\lambda$

c = speed of light, f = frequency, λ = wavelength.

Formula (E_0–B_0 relation): $B_0 = \dfrac{E_0}{c}$

E_0 = peak electric field, B_0 = peak magnetic field, c = speed of light.

Formula (Intensity): $I = \dfrac{1}{2}\varepsilon_0 c E_0^2$

I = intensity (average power per unit area), ε_0 = permittivity of free space, c = speed of light, E_0 = peak electric field.

Formula (Average energy density – electric): $\langle u_E \rangle = \dfrac{1}{4}\varepsilon_0 E_0^2$

$\langle u_E \rangle$ = time-averaged electric energy density; equal in magnitude to the average magnetic energy density.

Formula (Average energy density – magnetic): $\langle u_B \rangle = \dfrac{B_0^2}{4\mu_0} = \langle u_E \rangle$

$\langle u_B \rangle$ = time-averaged magnetic energy density, μ_0 = permeability of free space.

Formula (Total average energy density): $\langle u \rangle = \langle u_E \rangle + \langle u_B \rangle = \dfrac{1}{2}\varepsilon_0 E_0^2$

$\langle u \rangle$ = total time-averaged energy density of the EM wave (electric + magnetic contributions, which are equal).

Tips, Tricks & Hacks

- Displacement current $I_d = \varepsilon_0 \dfrac{d\phi_E}{dt}$ was Maxwell's fix so that Ampere's law works even inside a charging capacitor's gap, where there's no conduction current but the field is still changing.
- Total current (conduction + displacement) is continuous around any loop — inside a capacitor gap, I_d exactly equals the conduction current in the wires, so charge is never "lost".
- In EM waves, $\vec{E}$, $\vec{B}$, and the direction of propagation are mutually perpendicular, and E, B oscillate *in phase* with each other (not 90° out of phase — a frequent misconception).
- Speed of EM wave in vacuum: $c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}} = \dfrac{E_0}{B_0}$ — this ratio E_0/B_0 equalling c is a favourite one-liner to test.
- Order of the EM spectrum by increasing frequency (decreasing wavelength): radio → microwave → infrared → visible → UV → X-ray → gamma ray — memorise this sequence, it's tested almost every year (uses/production/detection of each).
- All EM waves travel at speed c in vacuum regardless of frequency — what changes across the spectrum is wavelength/frequency and the method of production/detection, not the speed.
- Energy in an EM wave is carried equally by the electric and magnetic fields (average energy density of each is equal) — useful for intensity-related numericals.